

2(a)	two homologous chromosomes circled ;	1
2(b)(i)	<i>any three from:</i> 1 same, size / length / number of (kilo)base pairs ; 2 same shape ; 3 same banding pattern ; 4 same (order of), genes / loci ; 5 they, pair up / form a bivalent, in, prophase 1 / meiosis 1 ;	3
2(b)(ii)	20 ;	1
2(c)(i)	<i>any two from:</i> 1 <i>ref. to</i> <u>reduction</u> division ; 2 halves the chromosome number ; 3 (so) fertilisation / fusion of gametes, gives, diploid / $2n$ / 40 ; 4 prevents chromosome <u>number</u> doubling (each generation) ;	2
2(c)(ii)	1 (by chance) both gametes (may) have recessive allele ; 2 zygote / offspring, has, pair / two / homozygous, <u>recessive</u> alleles ;	2
2(d)(i)	<i>any two from:</i> 1 (causes a) STOP codon ; 2 does not code for <u>amino acid</u> / no <u>amino acid</u> added to chain ; 3 stops <u>translation</u> OR rest of / later, codons / triplets / mRNA / sequence, not <u>translated</u> ;	2
2(d)(ii)	<i>parental genotypes</i> Aa A AA normal X and a Aa normal Aa A (Aa) (normal) a ; aa ; woolly ; <i>gametes</i> <i>offspring genotypes</i> <i>offspring phenotypes</i>	3